Fatemeh Shahrokhshahi

Istanbul, Turkey • ftm.shahrokhshahi@gmail.com • Homepage • LinkedIn • GitHub • Google Scholar

Research Interests

AI reliability: understanding, predicting, and preventing failures in large language models through reasoning analysis, targeted interventions, and interpretability.

Education

Master of Computer Engineering *Istanbul Aydin University, Istanbul, Turkey | GPA: 4.0/4.0* Sep 2024 - Present
Bachelor of Robotic Engineering *Shahrood University of Technology, Shahrood, Iran | GPA: 2.62/4.0* Sep 2015 - Feb 2022

Publications

Can We Predict LLM Reasoning Failures? Structural Predictability of Modal and Conditional Inference Errors
KI 2026: 49th German Conference on Artificial Intelligence, Bremen, Germany | Published Aug 2026
- Predicted per-instance LLM failures from input-side structural features (3,776 instances, nine models, three prompting strategies; AUC-ROC 0.69–0.93), showing that targeted prompting and output-side reliability mechanisms are complementary.

Towards Logical Reasoning by Design: Encoding Inference Rules in MeMo's Associative Memory Jun 2026
MeMo Workshop on Mechanistic Interpretability & Neuro-symbolic Approaches, University of Rome Tor Vergata | Accepted
- Encoded explicit inference rules in MeMo's Correlation Matrix Memory to build logical reasoning into the architecture.

LogiCue: Targeted Prompting for Improved Modal and Conditional Reasoning in LLMs Dec 2025
Procedia Computer Science, Elsevier (ACLing 2025) | Published
- Designed a pattern-specific prompting framework for seven modal inference patterns, evaluated across nine models with statistical validation ($p<0.01$).

Research Projects

Coefficient Sensitivity and Domain Robustness of Cross-Lingual Steering *(Pilot)* *Sep 2026*
Tested whether DiffMean-style activation steering (Llama-3.2-1B) stays robust under domain shift; found a narrow, sharply bounded effective coefficient window consistent across domains. [GitHub]

Balanced K-Means for Domain Discovery in LMs *Feb – May 2025*
Applied balanced k-means clustering to C4 to discover domains without general-domain dominance.

Intelligent Academic Retrieval System (RAG) *Nov 2024 – Feb 2025*
Built a RAG pipeline over research papers; embedding-based retrieval outperformed keyword search. [GitHub]

Academic Service

- Reviewer, NeurIPS 2026 Workshop: Who Verifies the Agents?, Sydney, Australia Sep 2026
- Program Committee Member (Reviewer), EITS 2026, Spain Apr 2026
- Presenter, ACLing 2025, British University in Dubai, UAE Dec 2025

Technical Skills

Programming: Python, SQL, C, Java, FastAPI
ML & NLP: PyTorch, Hugging Face Transformers, TensorFlow, LangChain, activation steering, fine-tuning, RAG
Research: Experimental design, statistical analysis, Git, AWS/GCP
Languages: Persian (native), English (C1, IELTS 7.0), French (A2), Turkish (basic)